B16 Takedown

Project: Spatial_Perturb
Author: Alexander Monell
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WEDNESDAY, 3/25/2026

- Target tissues: Spleen, dLN, tumor
- Buffers (For 10 mice)

3_26_Takedown_1.pdf

- Morning: Add 1 ml RP5 to wells of collection plate labeled with Spleen, dLN, tumor.
- Put collagenase into the water bath.
- Go down to mouse room and extract tissues.
- Come upstairs and add collagenase to the collagenase

B16 Tumor processing protocol obtained from Nicole. Thanks Nicole

- Tumor tissue dissociation
- Use scissors and chop tumor into small pieces inside 50ml falcon turned on its side (should look like mush).
- Add 10ml collagenase solution, washing scissors off and tissue down the side of the tube.
- Incubate for 30 min @ 37°C in shaking incubator.
- Pour through cell strainer into a new 50mL conical, mash through cell strainer, and add 10mL RP5 (RPMI +5% serum) to wash cells down.
- Put tubes on ice while processing other tumors.
- Before spinning, add 2mL 1x ACK to each tube, and vortex each tube for 30 seconds to further disassociate lymphocytes from tumor cells.
- Spin down at 1600 rpm for 6 min.

Percoll

- Make sure you see a pellet, then pour supernatant out.
- Resuspend each sample in 5 mL 44% percoll solution, make sure pellet is well resuspended, and transfer to a 15ml round bottom tube (bacteria culturing tube).
- Place glass pasteur pipette in conical and add 2 mL 67% percol solution to top of pipette to make a bottom layer. Don't disturb after adding!
- Spin at 21°C at 2000 rpm for 20 min, set acceleration to 3 and brake to 0
- Check for an interphase between the two layers and transfer it to 15 mL conical (usually about 2 mL).
- Fill to top with RP5 then spin at 1600 rpm for 4 min.

Ficoll

- Make sure you see a pellet, then pour supernatant out.

- Resuspend each sample in 5 mL RP5, place glass pasteur pipette in conical and add 2 mL Ficoll (Corning Lymphocyte Separation Media MT25072CV) to top of pipette to make a bottom layer. Don't disturb after adding!
- Spin at 10°C at 2000 rpm for 15 min, set acceleration to 9 and brake to 9
- Check for an interphase between the two layers and transfer it to 15 mL conical (usually about 2 mL).
- Fill to top with RP5 then spin at 1600 rpm for 4 min.
- Resuspend pellet in 1mL RP5 – cells are now ready to transfer to plates for staining.

B16_tumor_processing_protocol.docx

Staining

Whole panel

	Fluorochrome	Filter	Antibody	Dilution	E	F
1	FITC/GFP/YFP	488_530_30	GFP	---	X	
2	PerCP-Cy5.5	488_675_20 or 695_40	CD45.2	200	X	
3	PE, MitoOrange	561_582_15			X	
4	PE-TR, 594, RFP	561_610_20	mCherry	---		
5	PE-Cy5.5	561_670_14 or 710_50				
6	PE-Cy7	561_780_60	PD1	200	X	
7	Pac Blue/BV421	405_450_40	TCF	200	X	
8	BV510/ametrine	405_525_50	CD45.1	200	X	
9	BV605	405_610_20	CD8	200	X	
10	BV650	405_660_20				
11	BV711	405_710_50				
12	BV786	405_780_60	Tim3	200	X	
13	APC/AxF647/AxF660	628_670_40	Tox	50	X	
14	AxF700	628_730_45				
15	APC-Cy7/ef780	628_780_60	live/dead stain	500	X	

Unstimmed

- Count and plate cells.
- 50 ul FC block for 10 min on ice.
- Make SURFACE staining cocktail in FACS buffer - 100 ul per well.
    - Since I have 10 mice - 10 tumors, 10 dLN, 10 spleen

  In 3.2 ml FACS:

Surface stain

	Fluorochrome	Filter	Antibody	Dilution	total ul	F
1	PerCP-Cy5.5	488_675_20 or 695_40	CD45.2	200	16	
2	PE-Cy7	561_780_60	PD1	200	16	
3	BV510/ametrine	405_525_50	CD45.1	200	16	
4	BV605	405_610_20	CD8	200	16	
5	BV786	405_780_60	Tim3	200	16	
6	APC-Cy7/ef780	628_780_60	live/dead stain	500	6.4	

- Spin the plate with FC down at 2000rpm for 1 min
- Flick off the supernatant into the sink
- Add 100 ul of the antibody stain per well.
- Put on ice in the dark for 15 min.
- Add 150 ul FACS per well.
- Spin down at 2000 rpm for 1 min.
- Flick off the supernatant into the sink
- Resuspend in 100 ul of Biolegend 4% PFA in PBS
- Incubate at RT for 20 min in the dark.
- During incubation - make 100 ul per well of diluted ebio concentrate
    - For this, do 1 ml concentrate and 3 ml dilutent for a total of 4 ml.
- Add 150 ul FACS to wells when incubation is done.
- Spin 2000 rpm for 1 min.
- Flick off supernatant
- Resuspend cells in 100ul of the diluted concentrate.
- Incubate at RT for 20 min in the dark.
- During incubation - make intracellular stain.
    - First, do 1 ml Perm Buffer to 9 ml H2O for a buffer MM
    - Take 3.2 ml for stain in 5 ml eppie.

      Add the following:

Whole panel1

	Fluorochrome	Filter	Antibody	Dilution	total ul	F
1	Pac Blue/BV421	405_450_40	TCF	200	16	
2	APC/AxF647/ AxF660	628_670_40	Tox	50	64	

- After incubation, add 150 ul of the buffer MM to each well.
- Spin 2000 rpm for 1 min
- Flick off supernatant
- Resuspend cells in 100 ul antibody mix and incubate at RT for 30 min in the dark.
    - During incubation, make single color controls.
- After stain, add 150 ul of the buffer MM per well.
- Spin 2000 rpm for 1 min.
- flick
- Resuspend in 100 ul of FACS buffer and move to FACS tubes.

For flow, use the same voltages from Nancy's experiment while setting voltages on unstained.